# Exercises: Convex Optimization Problems

Mathematical Foundations for Machine Learning

## Terminology and Conditions for Convex Optimization

**Exercise 1.1** [Easy]
Consider the optimization problem

$$\min_{x\in\mathbb{R}^2} \; x_1^2 + x_2^2 \quad \text{s.t.} \quad x_1 + x_2 \le 3, \quad x_1 \ge 0, \quad x_2 \ge 0.$$

**(a)** Identify the objective function, the inequality constraint functions, and the feasible set.

**(b)** Is this a convex optimization problem in standard form? Verify each of the three requirements (convex objective, convex inequality constraints, linear equality constraints).

**(c)** Is the feasible set convex? Justify your answer.

**Exercise 1.2** [Easy]
For each of the following optimization problems, determine whether it is a convex optimization problem in standard form. If not, state which requirement is violated.

**(a)** $\min_{x\in\mathbb{R}} \; e^{-x} \quad \text{s.t.} \quad x^2 \le 1$

**(b)** $\min_{x\in\mathbb{R}^2} \; x_1^2 + x_2^2 \quad \text{s.t.} \quad x_1^2 + x_2^2 = 1$

**(c)** $\min_{x\in\mathbb{R}^2} \; x_1 + x_2 \quad \text{s.t.} \quad x_1^2 + x_2^2 \le 4, \quad 2x_1 - x_2 = 1$

**Exercise 1.3** [Medium]
Consider the problem

it is standard convex but not gradient in origin: 1D -> |x|, general grad: 0/0so ^2 is equ

$$\min_{x\in\mathbb{R}^n} \; \|x\|_2 \quad \text{s.t.} \quad Ax = b,$$

where $A \in \mathbb{R}^{m\times n}$ and $b \in \mathbb{R}^m$.

**(a)** This problem is not in standard convex form. Why not? (Hint: consider differentiability at the origin.)

**(b)** Show that this problem is equivalent to the convex optimization problem

$$\min_{x\in\mathbb{R}^n} \; \|x\|_2^2 \quad \text{s.t.} \quad Ax = b.$$

That is, both problems have the same set of optimal solutions.

**(c)** Verify that this reformulated problem is a convex optimization problem in standard form. Identify the problem class (LP, QP, QCQP, or SOCP).

**Exercise 1.4** [Medium]
Consider the problem of minimizing the maximum component of a vector subject to linear constraints:

$$\min_{x\in\mathbb{R}^n} \; \max_{i=1,\dots,n} x_i \quad \text{s.t.} \quad Ax = b.$$

**(a)** Is this a convex optimization problem in standard form? Justify.

**(b)** Reformulate it as an equivalent **linear program** by introducing an auxiliary variable $t \in \mathbb{R}$. Write the LP in standard form.

**(c)** What general principle does this illustrate about equivalent convex problems?

min of piecewise-linearconvex function into LP:min

## Local Optimality Implies Global Optimality

**Exercise 2.1** [Easy]
Let $\Omega \subseteq \mathbb{R}^n$ be a convex set and $f : \Omega \to \mathbb{R}$ be a convex function. Suppose $x^* \in \Omega$ is a local minimizer, i.e., there exists $R > 0$ such that $f(x^*) \leq f(z)$ for all feasible $z$ with $\|z - x^*\| \leq R$.

Prove that $x^*$ is a global minimizer: $f(x^*) \leq f(y)$ for all $y \in \Omega$.

**Exercise 2.2** [Medium]
global minimum: f(x) >= f(x*).

**(a)** Let $f : \mathbb{R}^n \to \mathbb{R}$ be strictly convex. Prove that if a global minimizer $x^*$ exists, then it is unique.

**(b)** Give an example of a strictly convex function $f : \mathbb{R} \to \mathbb{R}$ that does *not* attain its global minimum. What additional condition guarantees existence?

**(c)** Show that if $f$ is strongly convex with parameter $m > 0$, then $f$ is coercive, i.e., $f(x) \to \infty$ as $\|x\| \to \infty$. Conclude that a strongly convex, lower semi-continuous function on a closed convex domain always has a unique global minimizer.

**Exercise 2.3** [Hard]
convex -> all critical points global mins (one after the other)strictly -> at most one C

Let $f : \Omega \to \mathbb{R}$ be convex and continuously differentiable on a convex set $\Omega$, and denote by $X \subseteq \Omega$ the feasible set.

A convex function implies continuity on interior points, not in boundary ->

we could have a convex funct. defined in two regions

**(a)** Prove that $x^* \in X$ is optimal if and only if

$$\langle \nabla f(x^*), y - x^* \rangle \geq 0, \quad \forall y \in X.$$

(Hint: use the first-order characterization of convexity for one direction, and a directional argument for the other.)

**(b)** Suppose now $X = \{x : Ax = b\}$. Show that the optimality condition reduces to: there exists $\nu \in \mathbb{R}^s$ such that $\nabla f(x^*) + A^T \nu = 0$.

(Hint: every feasible direction is of the form $y - x^* = v$ where $v \in \ker(A)$.)

## Linear and Quadratic Programs

**Exercise 3.1** [Easy]
Classify each of the following problems as an LP, QP, QCQP, SOCP, or none of these. Justify briefly.

**(a)** $\min_{x \in \mathbb{R}^3} \; 2x_1 - x_2 + 4x_3 \quad \text{s.t.} \quad x_1 + x_2 + x_3 = 1, \; x_i \geq 0$

**(b)** $\min_{x \in \mathbb{R}^2} \; x_1^2 + 3x_2^2 + 2x_1x_2 \quad \text{s.t.} \quad x_1 + x_2 \leq 5$

**(c)** $\min_{x \in \mathbb{R}^2} \; x_1 + x_2 \quad \text{s.t.} \quad x_1^2 + x_2^2 \leq 4, \; x_1^2 - x_2 \leq 0$

**Exercise 3.2** [Medium]
A company produces $n$ products. The profit per unit of product $i$ is $c_i > 0$. Production of product $i$ requires $a_{ji}$ units of resource $j$, and there are $b_j$ units of resource $j$ available ($j = 1, \ldots, m$). Let $x_i$ be the quantity produced of product $i$.

**(a)** Formulate the profit maximization problem as a linear program.

**(b)** Convert this to a minimization LP in standard form as defined in the lecture (with equality constraints and non-negativity).

**(c)** Now suppose the profit per unit is uncertain: $c_i$ is a random variable with mean $\bar{c}_i$ and the cost vector $c$ has covariance $\Sigma \succ 0$. The company wants to maximize the risk-adjusted profit $\mathbb{E}[c^T x] - \gamma \operatorname{Var}(c^T x)$ for some $\gamma > 0$. Formulate this as a QP.

**Exercise 3.3** [Medium]
Consider Ridge Regression: given data $X \in \mathbb{R}^{N\times D}$, targets $y \in \mathbb{R}^N$, and regularization parameter $\lambda > 0$:

$$\min_{\theta\in\mathbb{R}^D} \|X\theta - y\|_2^2 + \lambda\|\theta\|_2^2.$$

**(a)** Show this is a QP by identifying the matrix $P$, the vector $q$, and the constant $c$ in the standard form $\min \frac{1}{2}\theta^T P\theta + q^T\theta + c$.

**(b)** Prove that $P \succ 0$ for any $\lambda > 0$ and any $X$, and conclude that the problem is strictly convex with a unique minimizer.

**(c)** Derive the closed-form solution by setting the gradient to zero.

**Exercise 3.4** [Hard]
Consider the Chebyshev center problem: find the center $x_c$ and largest radius $R$ of a Euclidean ball $B(x_c, R) = \{x_c + u : \|u\|_2 \le R\}$ contained in a polyhedron $P = \{x \in \mathbb{R}^n : \langle a_i, x\rangle \le b_i,\ i = 1, \dots, r\}$.

**(a)** Show that the constraint $B(x_c, R) \subseteq \{x : \langle a_i, x\rangle \le b_i\}$ can be written as

$$\langle a_i, x_c\rangle + R\|a_i\|_2 \le b_i.$$

(Hint: the ball lies in the half-space iff $\sup\{\langle a_i, x_c + u\rangle : \|u\| \le R\} \le b_i$.)

**(b)** Formulate the full optimization problem and show it is a linear program in the variables $(x_c, R)$.

**(c)** Explain why this is a convex optimization problem even though we are *maximizing* $R$.

## QCQP and SOCP

**Exercise 4.1** [Medium]
Consider the robust linear program: given uncertainty sets $\mathcal{E}_i = \{\bar{a}_i + P_i u : \|u\|_2 \le 1\}$ with $P_i \in \mathbb{S}^n_+$, we want

$$\min_{x\in\mathbb{R}^n} c^T x \quad \text{s.t.} \quad \langle a_i, x\rangle \le b_i, \quad \forall\, a_i \in \mathcal{E}_i,\ i = 1, \dots, r.$$

**(a)** Show that the robust constraint $\langle a_i, x\rangle \le b_i$ for all $a_i \in \mathcal{E}_i$ is equivalent to

$$\langle \bar{a}_i, x\rangle + \|P_i x\|_2 \le b_i.$$

**(b)** Is this constraint linear, quadratic, or a second-order cone constraint? What problem class does the full robust LP belong to?

**Exercise 4.2** [Hard]
Show the inclusion chain LP $\subset$ QP $\subset$ QCQP $\subset$ SOCP by answering:

**(a)** Explain why every LP is a QP. What is the matrix $P$ in the quadratic objective?

**(b)** Explain why every QP is a QCQP.

**(c)** Show that a convex quadratic constraint $\frac{1}{2}x^T Px + q^T x + c \le 0$ with $P \in \mathbb{S}^n_+$ can be written as a second-order cone constraint. (Hint: use the Cholesky factorization $P = LL^T$ and complete the square.)

**(d)** For each inclusion, give a concrete example showing the inclusion is strict (i.e., a problem in the larger class that is *not* in the smaller class).

## Mixed Challenging Exercises

**Exercise 5.1** **[Hard]**
Consider the problem of finding the point in a convex polyhedron $P = \{x \in \mathbb{R}^n : Ax \leq b\}$ closest to a given point $p \in \mathbb{R}^n$:

$$\min_{x \in \mathbb{R}^n} \|x - p\|_2^2 \quad \text{s.t.} \quad Ax \leq b.$$

**(a)** Classify this as an LP, QP, QCQP, or SOCP. Identify $P$, $q$, $c$ in standard quadratic form.

**(b)** Prove that this problem always has a unique solution (state clearly which theorems you use).

**(c)** Use the optimality criterion for differentiable convex functions: show that $x^*$ is optimal if and only if $\langle x^* - p,\, y - x^* \rangle \geq 0$ for all $y \in P$. Give a geometric interpretation of this condition.

**(d)** In the special case $P = \{x : x_1 + x_2 \geq 4,\ 2x_1 + x_2 \geq 5\}$ and $p = (0, 0)^T$, use the KKT conditions to find the closest point to the origin.

**Exercise 5.2** **[Hard]**
In training a neural network, we minimize a loss $\ell(\theta)$ over parameters $\theta \in \mathbb{R}^D$ subject to a norm constraint to prevent overfitting:

$$\min_{\theta \in \mathbb{R}^D} \ell(\theta) \quad \text{s.t.} \quad \|\theta\|_2^2 \leq C.$$

**(a)** Suppose $\ell$ is convex. Is this a convex optimization problem? Which problem class (LP, QP, QCQP, SOCP, general convex) does it belong to? Justify.

**(b)** Suppose furthermore that $\ell$ is strictly convex. Using the results on convex optimization, what can you conclude about the minimizer?

**(c)** In practice, neural network losses are *non-convex*. Explain why the theorem "local min $\Rightarrow$ global min" does *not* apply, and give a simple one-dimensional example of a non-convex function with a local minimum that is not global.

**(d)** Despite non-convexity, practitioners use gradient descent and often find good solutions. Suggest one reason, from an optimization perspective, why this might work. (1–2 sentences suffice.)